Roll No.: 24005

Amrita Vishwa Vidyapeetham

Integrated MSc Degree Examination – June 2025

Second Semester

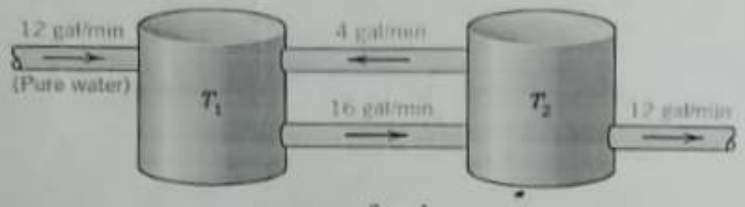
Common For Mathematics and Physics

24MAT 121 Ordinary Differential Equations

Time : Three hours | Maximum : 100 Marks

Answer All Questions

Qn No.	Questions	Marks	CO	Blooms Taxonomy level
1	a) A colony of bacteria is growing exponentially. At time $t = 0$, it has 10 bacteria in it, and at time $t = 4$ it has 2000. At what time will it have 100,000 bacteria? b) Check for exactness and then solve the differential equation: $e^{2x}(2cosydx - sinydy) = 0$. c) Verify that $y = (c + x)e^x$ is a solution of the differential equation $\frac{dy}{dx} = y + e^x$.	4 4 2	CO1	L_3
2	a) What do you mean by linear independence and dependence of solution? Check whether the functions x, x^2, and x^3 are linearly independent or dependent? Justify your answer. b) Use method of reduction of order to find a basis of solution of the differential equation $x^2y'' - 3xy' + 4y = 0$, if $y_1 = x^2$ is a solution.	5 5	CO2	L_3
3	a) Find the general solution of $y'' - 5y' + 7y = 0$. b) Using proper method solve the differential equation $x^2y'' - 2xy' + 5y = 0$.	5 5	CO2	L_3
4	a) Find the Laplace transform of i) $e^{-3t}cost$ ii) $tsin2t$ iii) $5t^4 - 20e^{8t}$. b) Find the inverse Laplace transform of $\frac{e^{-s}}{s^2+3s+2}$ using partial fraction method.	5 5	CO4	L_4
5	a) Using proper results of Laplace transform, solve the differential equation $y'' + y = sin\,t, y(0) = 0, y'(0) = 0$. b) Find the inverse Laplace transform of $ln\left(\frac{s+2}{s+5}\right)$.	6 4	CO5	L_4
6	By the method of determining undetermined coefficients, find a solution of the ordinary differential equation $y'' - 2y' + 5y = sin3x$.	10	CO2	L_3

7	By the method of variation of parameters, find a solution of the ordinary differential equation $x^2y'' - 2xy' + 2y = x^3 sinx$.	10	CO2	L_3
8	Each of the two tanks contains 200 gal of water, in which initially 100 pounds (Tank T_1) and 200 pounds (Tank T_2) of fertilizer are dissolved. The inflow, circulation, and outflow are shown in the figure. The mixture is kept uniform by stirring. Find the fertilizer contents in both the tanks at any time t.  system of tanks	10	CO2	L_3
9.	Solve the initial value problem $y''' - y'' + 100y' - 100y = 0$, with $y(0) = 4, y'(0) = 11, y''(0) = -299$, by using method of determining undetermined coefficients.	10	CO3	L_3
10	a) By the method of power series solve the differential equation $y'' + y = 0$. b) Find the radius of convergence of the series $\sum_{m=1}^{\infty} \frac{4^m}{m(m+1)} x^m$.	7 3	CO3	L_3

L_1-Remember, L_2-Understand, L_3-Apply, L_4-Analyze, L_5-Evaluate, L_6-Create